Fake News/Misinformation Detection (NLP)

Adrian John C. Alfonso
Computer Science Undergraduate
Mapaumanhin St. Manibaug Paralaya
Porac, Pampanga
+63 9476175343
adrianjohnalfonso47@gmail.com

Laurenzo S. Centeno
Computer Science Undergraduate
Jesus Street, Pulung Bulu
Angeles City, Pampanga
+63 926 022 8287
centenoenzo054@gmail.com

Seane Karl S. Garcia
Computer Science Undergraduate
San Roque St., Sta Cruz
Porac, Pampanga
+63 9396039256
grcseane@gmail.com

John Michael Y. Supan
Computer Science Undergraduate
Anita Road, Essel Park Subdivision
City of San Fernando, Pampanga
+63 9610042777
supanjohn13@gmail.com

ABSTRACT

The rapid spread of misinformation on digital platforms is still prevalent and has increased the need for automated systems capable of detecting fraudulent news efficiently. This study will develop a machine learning–based system that will classify textual claims as either real or fake using Natural Language Processing (NLP) techniques. The proposed approach will utilize a compact transformer model, DistilBERT, to analyze contextual relationships between words and identify linguistic patterns associated with misinformation. The system will be trained and evaluated using the LIAR dataset, a publicly available collection of fact-checked political statements labeled according to their truthfulness. The model's performance will be evaluated using metrics such as Macro-F1 score to measure classification effectiveness. The expected outcome of this research is a lightweight and efficient fake news detection system that will assist in identifying misleading information in textual content.

CCS Concepts

• Computing methodologies → Natural language processing

• Computing methodologies → Machine learning

• Information systems → Data mining

Keywords

*Fake News Detection; Natural Language Processing; DistilBERT; Transformer Models; Text Classificatio*n

1. INTRODUCTION

The rapid growth of digital media and online social networks has significantly increased the spread of misinformation and fake news. False information can easily reach a large audience within a short period of time, which may influence public opinion, political decisions, and social stability. As a result, detecting fake news has become an important research problem in the field of natural language processing and machine learning [1].

Traditional methods of identifying fake news often rely on manual fact-checking, which can be time-consuming and difficult to scale as the amount of online content continues to grow. Automated fake news detection systems have therefore been developed to analyze textual information and classify whether a piece of content is real or misleading [4].

Recent studies have explored the use of machine learning and deep learning models to automatically detect misinformation. In particular, transformer-based language models have demonstrated strong performance in various natural language processing tasks due to their ability to capture contextual relationships within text. [3] proposed FakeBERT, a deep learning model that combines BERT with convolutional neural networks to improve fake news detection accuracy. Inspired by these approaches, this study will develop a fake news detection system using a transformer-based model, DistilBERT, which will classify textual claims as either real or fake.

The system will be trained and evaluated using the LIAR dataset, a publicly available benchmark dataset consisting of fact-checked political statements collected from the PolitiFact website. The dataset contains over 12,000 labeled statements and is widely used for research in fake news detection and misinformation classification tasks [8].

The goal of this research is to design an efficient and lightweight fake news detection system that can analyze textual claims and assist in identifying misleading information in digital media.

1.1 Success Metrics

The performance of the proposed fake news detection system will be evaluated using a multi-faceted approach to ensure robust and

unbiased results. Central to this evaluation is the Macro F1-score, which serves as the primary metric by measuring the balanced performance between the "fake" and "real" classes. By utilizing the harmonic average of precision and recall, the Macro F1-score ensures that the model evaluation is not biased toward the majority class, a factor that is particularly critical in fake news detection where datasets are often heavily imbalanced. In addition to this, overall classification accuracy will be monitored to assess the general correctness of predictions across the entire dataset. To provide a more granular view of model performance, precision and recall will also be analyzed individually, evaluating the correctness and completeness of predictions for each specific category. Finally, the system will measure Reinforcement Learning (RL) performance during the optimization phase, specifically tracking the iterative improvement in the Macro F1-score following the application of RL-based threshold optimization.

1.2 Constraints

The proposed system will be developed under several operational and technical constraints. Primarily, the model training and evaluation pipeline must be completed within a one-week timeframe on a single mid-range GPU, as specified in the project requirements. To meet the core project objectives, the architecture (CNN), Natural Language Processing (NLP), and Reinforcement Learning (RL) components, all of which must run reproducibly using the provided scripts from the designated GitHub repository.

The system's performance is heavily contingent on the quality and scope of the dataset; specifically, the model relies on the LIAR dataset, which consists primarily of short political statements. This specific focus may limit the model's generalization to other domains of misinformation or longer-form articles. Furthermore, the system is strictly limited to text-based fake news detection and does not analyze multimedia content such as images or videos, nor does it perform multi-class labeling, as it is designed solely for binary classification (real vs. fake). Finally, due to limited computational resources during development, the reinforcement learning approach remains a simplified implementation and is not fully optimized for all possible edge cases.

1.3 Problem Statement

Despite advancements in automated fact-checking, current models often struggle with the inherent ambiguity and brevity of short-form political statements, leading to skewed classification performance. The core challenge addressed by this research is the development of a balanced detection system that can effectively mitigate the trade-off between precision and recall in imbalanced datasets. Furthermore, there remains a technical gap in optimizing decision thresholds dynamically; thus, this study aims to determine how an integrated CNN, NLP, and Reinforcement Learning framework can be leveraged to improve classification robustness under significant computational and data constraints.

2. Scope and limitations

This study will focus on developing a text-based fake news detection system using a transformer-based model, specifically DistilBERT. The system will focus on text classification of news content using deep learning models. The project will implement both transformer-based NLP models and a CNN-based text classification model to analyze textual features associated with misinformation. The system will be designed with a reproducible machine learning pipeline and documented dataset provenance. Additionally, the system will only analyze textual information and will not evaluate other forms of media such as images, videos, or multimedia content commonly used in misinformation campaigns. The proposed model will also be constrained by the size and characteristics of the dataset used for training, which may affect its ability to generalize to unseen data.

3. Related Work

3.1 Fake News Detection Using BERT

The increasing spread of misinformation on online platforms has caused a need for systems capable of detecting fake news efficiently through automation. [2] investigated the use of machine learning and transformer-based language models for identifying fake news from textual content. Their study shows that transformer architectures, such as BERT, can effectively determine contextual relationships between words and extract meaningful linguistic features from news articles. By leveraging deep contextual representations, these models improve the ability to distinguish between legitimate information and misleading content. The findings highlight the effectiveness of transformer-based approaches in fake news detection tasks and support the use of compact transformer models such as DistilBERT for efficient text classification.

3.2 Fake News Detection Using Transformers

Recent studies have explored the use of transformer-based models for detecting misinformation in textual data. Kaliyar et al. (2021) proposed FakeBERT, a deep learning model that combines the Bidirectional Encoder Representations from Transformers (BERT) architecture with convolutional neural network layers for fake news detection. The model leverages BERT's bidirectional language representation to capture semantic relationships and contextual information within news text, enabling the system to better identify misleading or fabricated information. Experimental results demonstrated that the proposed approach outperformed several traditional machine learning methods, highlighting the effectiveness of transformer-based models in fake news classification tasks.

3.3 Fake News Detection Using Deep Learning

Deep learning has become widely used in fake news detection because it can learn patterns from large amounts of textual data. Al-Tai et al. (2023) reviewed different deep learning techniques used to detect misinformation and discussed models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and attention-based architectures for analyzing news text. These models use word embeddings to convert text into numerical representations so that the system can understand the meaning and context of words in news articles. The study shows that deep learning models can improve the accuracy of fake news detection by learning linguistic patterns from large datasets, making them useful tools for identifying misleading or false information.

3.4 Fake News Scanning Automation NLP/Deep Learning

Automated systems have been developed to detect the spread of misinformation on social media platforms. Leonardi et al. (2021) proposed a computational approach that analyzes linguistic features from social media posts to identify users who spread fake news. Their model uses transformer-based language representations to convert social media posts into sentence embeddings, which are then processed by a deep learning classifier. The study showed that combining natural language processing techniques with deep learning models can effectively classify fake news spreaders and improve misinformation detection accuracy. This demonstrates the potential of automated NLP-based systems in supporting fact-checking efforts and reducing the spread of misleading information online.

4. Dataset Plan

The dataset that will be used in this study is the LIAR dataset, a publicly available benchmark dataset for fake news detection research. The dataset contains approximately 12,800 short political statements collected from the PolitiFact fact-checking website. Each statement has been manually verified and labeled according to its credibility level.

The dataset includes six categories of truthfulness: pants-on-fire, false, barely-true, half-true, mostly-true, and true. In addition to the textual statement, the dataset also provides metadata such as the speaker, political party affiliation, subject, and context in which the statement was made.

The LIAR dataset was selected because it provides high-quality fact-checked data and is widely used as a benchmark dataset in fake news detection research. Using this dataset allows the proposed system to be evaluated against established research studies and ensures that the model is trained on reliable labeled data.

4.1 Data Acquisition and Pipeline

To ensure pipeline reliability and avoid external library dependencies, the data acquisition process was engineered to bypass third-party repositories like Hugging Face. The custom script, get_data.py, directly interfaces with the UC Santa Barbara repository to download the official LIAR dataset. This approach extracts raw TSV files and performs local directory cleaning, guaranteeing full reproducibility without committing large datasets to version control.

The automated preparation engine, process_dataset.py, transforms the raw data into a model-ready format. This involves assigning structured schema names, filtering null values, and standardizing text through case normalization and punctuation removal. Most importantly, the script maps the original six-point truth scale into a streamlined binary format (0 for Fake, 1 for Real) before exporting the finalized subsets as CSV files for training.

4.2. Exploratory Data Analysis

The EDA notebook (01_eda.ipynb) was developed to process these raw TSV files. The original six-point truth scale was collapsed into a strict binary classification target: 0 for Fake and 1 for Real. Missing text values were dropped, and sequence lengths were calculated to inform upcoming transformer tokenization limits.

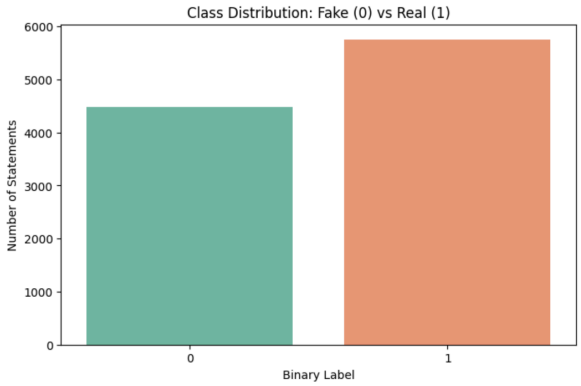

Figure 1: *The class distribution of the binary mapped LIAR dataset, showing the balance between Fake (0) and Real (1) statements.*

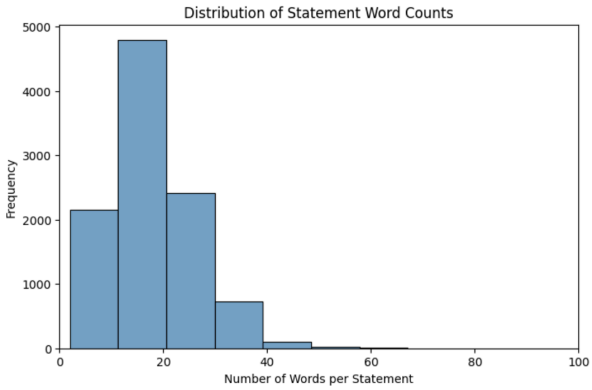

Figure 2: *Distribution of statement word counts, utilized to set the max_length parameter for the DistilBERT tokenizer.*

5. Proposed Method

5.1 Machine Learning and DL Prototyping

Before training resource-intensive neural networks, a traditional machine learning baseline was established using Scikit-Learn. The cleaned text was vectorized using TF-IDF and fitted to a Logistic Regression model to capture initial Accuracy and Macro F1 scores.

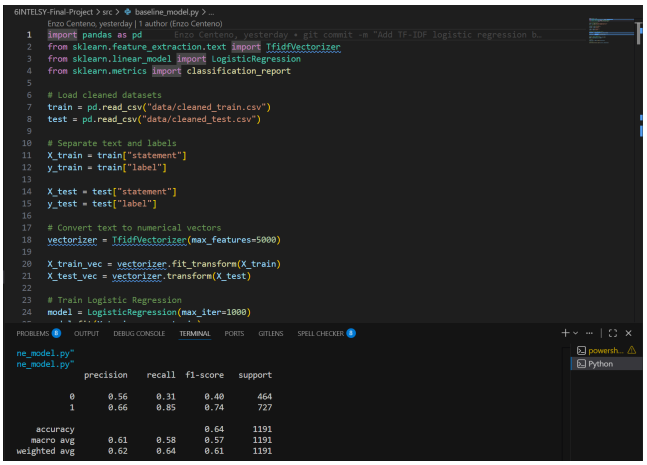

Figure 3: *Initial classification report for the TF-IDF and Logistic Regression baseline.*

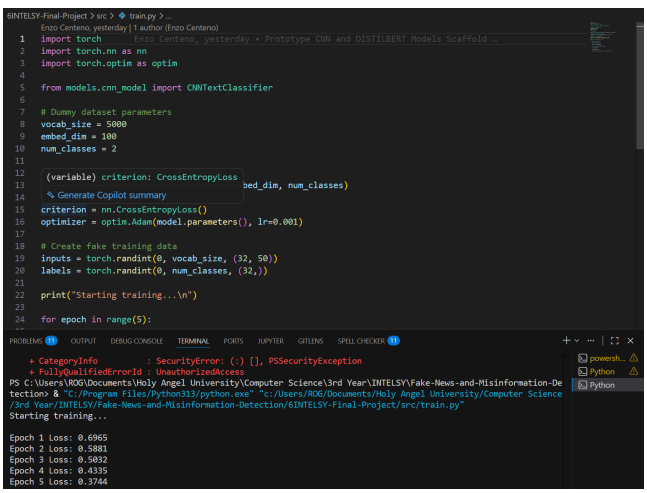

Figure 4: *Early training loss curve for the PyTorch 1D-CNN prototype, demonstrating successful gradient descent.*

Simultaneously, the core PyTorch architectures were prototyped in the src/models/ directory. A 1D-Convolutional Neural Network (CNN) was built using nn.Embedding and nn.Conv1d layers to capture local linguistic features. A preliminary training loop was executed to verify that the model compiles and effectively learns from the batches. Furthermore, the Hugging Face transformers library was used to scaffold the DistilBERT class, preparing the pipeline for full fine-tuning in the upcoming phase.

Reinforcement Learning Environment Stub

A custom environment was built using the gymnasium library to house the Reinforcement Learning agent. The environment defines the "action space" as the adjustment of the classification probability threshold, and the "reward" as the resulting Macro F1 score evaluated on the validation set. A preliminary script executing random threshold actions over 50 episodes was run to validate the environment mechanics and generate an early learning curve.

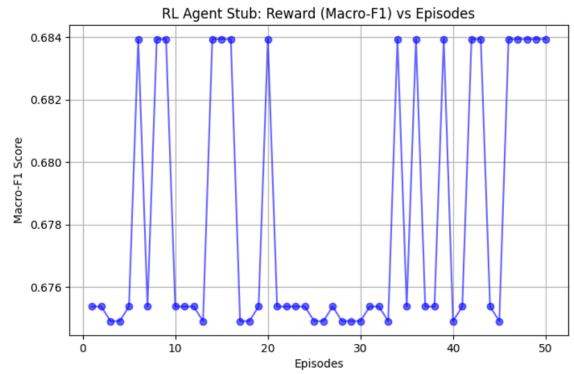

Figure 5: *Early, unoptimized learning curve of the RL agent interacting with the threshold environment.*

5.2 NLP Component

The core component of the system is a Natural Language Processing (NLP) model designed to classify textual posts as either real or fake. This study will utilize a compact transformer-based model, specifically DistilBERT. It is a lightweight version of the BERT architecture optimized for faster training and inference while also maintaining strong language understanding capabilities.

5.3 CNN Component

In addition to the transformer model, a Convolutional Neural Network (CNN) will be implemented as a baseline model for comparison. CNN models are commonly used in text classification tasks because they can extract meaningful features from word embeddings through convolutional filters.

5.4 Reinforcement Learning Component

A reinforcement learning component will be incorporated to optimize the decision threshold used for classification. Instead of relying on a fixed threshold, the reinforcement learning agent will adjust the threshold dynamically based on model performance during validation.

5.5 Minimum Viable Product

The minimum viable product of the proposed system will consist of a basic text classification interface that allows users to input a news statement or claim.

5.6 Ethical Considerations and Risk Mitigation

The team has initiated an active Risk Register to track ethical concerns throughout the development lifecycle. To mitigate algorithmic bias, the model prioritizes the Macro F1 metric over simple accuracy; this ensures that the system performs equitably across both majority and minority classes, preventing the model

from ignoring underrepresented linguistic patterns. To address the risk of over-censorship and false positives, the Reinforcement Learning (RL) agent dynamically optimizes the decision threshold. This moving boundary reduces the likelihood of legitimate news being flagged as misinformation compared to a rigid 0.5 cutoff.

Furthermore, the project maintains Data Privacy and Provenance by strictly utilizing the public LIAR dataset, which consists of publicly made political statements. This eliminates the need for unauthorized web scraping and ensures no Protected Personally Identifiable Information (PII) is processed. Finally, to ensure Transparency, the system is designed as a "Human-in-the-loop" assistant rather than an autonomous censor; the output will be presented as a probability score to aid human fact-checkers, acknowledging that automated systems should support, not replace, human judgment in complex socio-political contexts.

6. Ethics Risk Register

The development of automated fake news detection systems introduces several ethical risks that must be considered during the design and deployment of the system. The following risks and mitigation strategies are identified:

6.1 Dataset Bias

The dataset used for training may contain biases toward certain topics, writing styles, or political viewpoints. This may cause the model to produce biased predictions. To mitigate this issue, the dataset will be analyzed and balanced when possible, and model performance will be evaluated across different categories of statements.

6.2 False Positives and Misclassification

The model may incorrectly classify legitimate news as fake or fail to detect certain misleading statements. To reduce this risk, the system will provide F probability scores rather than strict binary labels and will emphasize that the predictions are intended to assist users rather than replace human verification.

6.3 Misuse of the System

Users may rely entirely on automated predictions without critically evaluating the information. To address this concern, the system will include disclaimers indicating that the results are recommendations and should be verified using reliable sources.

7. Team Roles and Milestone

The project will be completed through collaboration among four team members, each responsible for specific components of the system.

7.1 Team Roles

- Seane Karl S. Garcia – Project Lead

  Responsible for coordinating the project, organizing documentation, and ensuring that all components of the proposal and final report are completed on time.

- John Micheal Y. Supan – Dataset Engineer

  Responsible for collecting, cleaning, and preprocessing the dataset used for training and evaluating the machine learning models.

- Laurenzo S. Centeno – NLP Developer

  Responsible for implementing the transformer-based fake news classification model using DistilBERT and conducting model training and evaluation.

- Adrian John C. Alfonso – Model Engineer

  Responsible for implementing the CNN baseline model and the reinforcement learning component for optimizing classification thresholds.

7.2 Project Milestones

- Week 1: Proposal preparation, dataset selection, and initial project planning.
- Week 2: Model implementation and training of the DistilBERT and CNN models.
- Week 3: Model evaluation, performance analysis, and preparation of the final documentation.

8. REFERENCES

[1] Al-Tai, M. H., Nema, B. M., and Al-Sherbaz, A. 2023. Deep Learning for Fake News Detection: Literature Review. Al-Mustansiriyah Journal of Science. https://mjs.uomustansiriyah.edu.iq/index.php/MJS/article/view/1292/674

[2] Anggrainingsih, R., Apriliyani, I., and others. 2021. Evaluating BERT-based Language Models for Detecting Fake News. International Journal of Machine Learning and Cybernetics. https://doi.org/10.1007/s41060-021-00302-z

[3] Kaliyar, R. K., Goswami, A., and Narang, P. 2021. FakeBERT: Fake News Detection in Social Media with a BERT-Based Deep Learning Approach. Multimedia Tools and Applications. https://doi.org/10.1007/s11042-020-10183-2

[4] Leonardi, S., Rizzo, G., and Morisio, M. 2021. Automated Classification of Fake News Spreaders to Break the Misinformation Chain. Information, 12(6), 248. https://doi.org/10.3390/info12060248

[5] Nasir, J. A., Khan, O. S., and Varlamis, I. 2021. Fake News Detection: A Hybrid CNN-RNN Based Deep Learning Approach. International Journal of Information Management Data Insights. https://doi.org/10.1016/j.jjimei.2020.100007

[6] Sanh, V., Debut, L., Chaumond, J., and Wolf, T. 2019.DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter. https://arxiv.org/abs/1910.01108

[7] Sutton, R. S. and Barto, A. G. 2021.Reinforcement Learning: An Introduction, 2nd ed. MIT Press, Cambridge, MA. https://incompleteideas.net/book/the-book-2nd.html

[8] Wang, W. Y. 2017. “Liar, Liar Pants on Fire”: A New Benchmark Dataset for Fake News Detection. Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL). https://aclanthology.org/P17-2067/